FPT UNIVERSITY - CAMPUS CAN THO

FPT UNIVERSITY

STAR-UP FINAL PROJECT

EXEPERIENTIAL ENTREPRENEURSHIP 2 (EXE201)

FALL 2026

HANDMADE PRODUCTS BRAND FROM RECYCLED PLASTIC CAPS

CAPCHAIN



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CHAPTER 1: OVERVIEW

1.1 GENERAL INFORMATION

1.1.1 Company Information

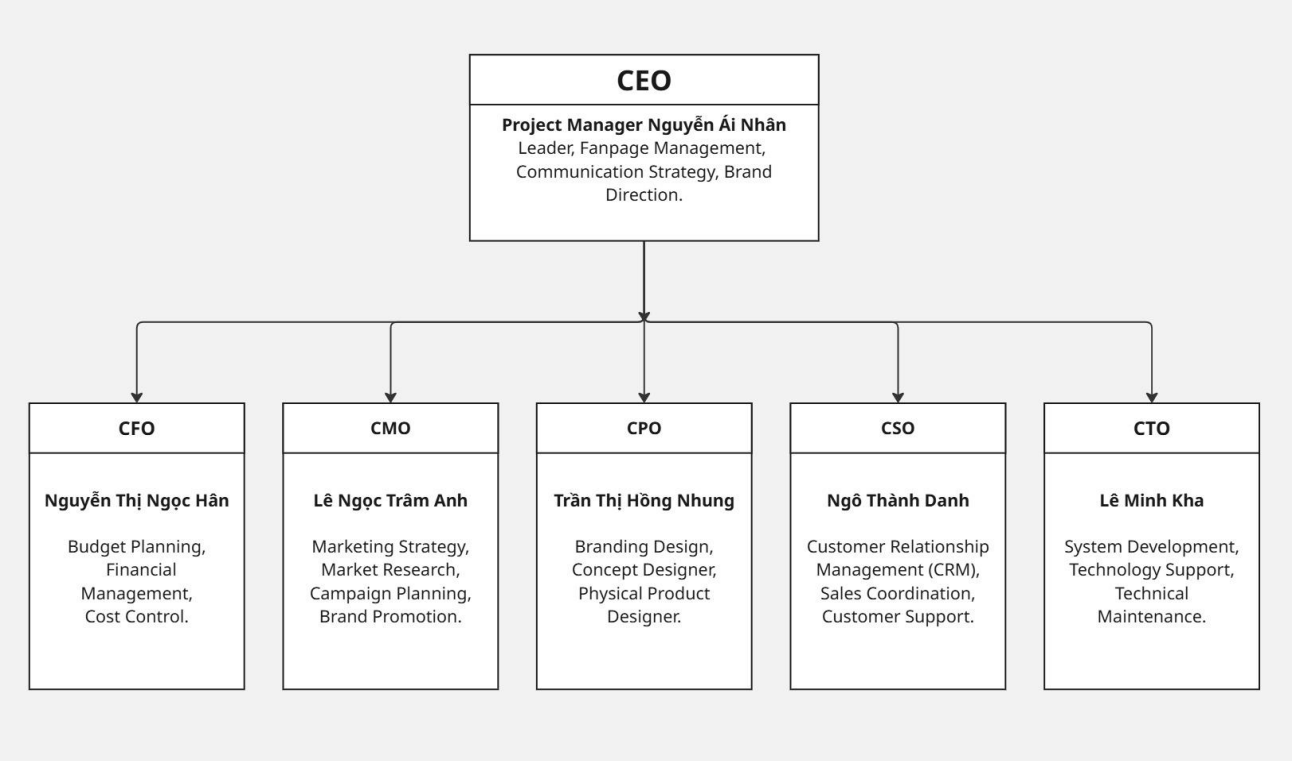
- Group's name: CapChain| Connecting Green Innovation
- Address: 600 Nguyen Van Cu street, Ninh Kieu ward, Can Tho city.
- Logo:



The logo is inspired by colorful bottle caps, creating artistic and stylish products, conveying a message of recycling things like plastic bottles or bottle caps, creating countless new colors and shapes from bottle caps.

- Fanpage: https://www.facebook.com/share/1AEtjhhn5x/?mibextid=wwXIfr
- Website: https://capchain.id.vn/
- Email: capchain101@gmail.com
- Established date: 12/05/2026

1.1.2 Company's Department



Roles of team members

Name	Position	Descriptions
Nguyễn Ái Nhân	CEO	Manages and coordinates all project activities.
Nguyễn Thị Ngọc Hân	CFO	Responsible for project finance, budgeting, and cost management.
Lê Ngọc Trâm Anh	CMO	Responsible for marketing strategy, market research, campaign planning, and brand promotion.
Trần Thị Hồng Nhung	CPO	Responsible for product design and brand identity development.
Ngô Thành Danh	CSO	Responsible for customer relationship management, sales coordination, customer support.
Lê Minh Kha	CTO	Responsible for technology, website system development, and technical support.

1.1.3 Implementation process

1.1.3.1 Project Construction and Product Development Phase (R&D & MVP)

Before launching CapChain products, the project began with market research to understand the needs and preferences of its target customers. The main target groups are university students, particularly Gen Z consumers, and people who are interested in eco-friendly lifestyles. The research identified a key customer pain point: recycled products available on the market are often perceived as rough in appearance, with limited aesthetic value and personalization. Therefore, CapChain aims to address this gap by developing handmade recycled products that combine environmental value with attractive and personalized designs.

Based on these findings, CapChain established several product criteria. The products should have a unique Marble Effect, created naturally from the combination of different recycled plastic colors. They should also provide good durability through the use of HDPE/PP plastic materials while maintaining affordable prices suitable for students, ranging from VND 25,000 to below VND 50,000. In addition to their functional and aesthetic value, the products are designed to communicate a strong environmental message and encourage customers to participate in a more sustainable lifestyle.

Following the market research stage, CapChain conducted research and development (R&D) to determine suitable production techniques. The project focused on heat-pressing techniques for recycled plastic bottle caps and manual cutting techniques for plastic bottles. Through this process, CapChain developed a handcrafted production method consisting of five main stages: collecting plastic caps and bottles, sorting them by color, shredding or grinding the materials, heat pressing them to create the characteristic Marble Effect, and finally molding and shaping the materials into the desired product designs.

The first Minimum Viable Product (MVP) was the CapChain Recycled Keychain collection. Several designs were tested to evaluate customer interest and product feasibility, including the Alphabet Collection, featuring letters from A to Z; the Nature Collection, featuring leaf and flower shapes; the Character Collection, featuring Chiikawa-inspired characters; and the Mini Art Collection, featuring simple

shapes such as hearts and stars. These initial products allowed the project to test both the production process and the appeal of different designs among potential customers.

After developing the initial keychain products, CapChain expanded its product line through MVP Version 2. This stage introduced additional products, including pocket-sized mini mirrors made from flattened bottle caps and mini plant pots made from recycled PET bottles. The expansion allowed CapChain to diversify its product portfolio while continuing to utilize recycled plastic materials.

CapChain focused on improving packaging and the overall user experience. Kraft tags were designed with environmental messages, such as “This product saved 5 plastic bottle caps from ending up in landfills,” to help customers understand the environmental impact of their purchases. The project also uses Kraft paper boxes and bags printed with organic ink, reinforcing the brand's commitment to environmentally responsible production and creating a consistent sustainable experience from the product itself to its packaging.

1.1.3.2 Product Refinement and Test Sales Phase

After the initial product testing stage, CapChain finalized three core product lines, including recycled plastic keychains, pocket-sized mini mirrors, and mini green plant pots designed for study desk decoration. The products were standardized to ensure practical use and durability, including water resistance, resistance to minor impacts, rust-proof stainless steel key rings, and smart drainage holes for the mini plant pots.

To introduce the products to customers, CapChain conducted social media campaigns from July 30 to August 5, 2026. The campaign included brand awareness teasers, the launch of the Marble Keychain Collection, and 45-second Reels featuring ASMR and timelapse videos of the production process. The Mini Plant Pot line was also introduced together with special product combos.

From August 7 to August 8, 2026, CapChain conducted direct test sales at a booth in Gamma Hall, FPT University Can Tho. The booth featured the “Trade Caps for Greenery” initiative, allowing students to exchange five bottle caps for a 10% discount voucher. Customers could also participate in a DIY corner where they could paint and decorate their own mini plant pots.

The test sales generated positive results, with over 100 keychains and 40 mini plant pots sold, while more than 500 plastic bottle caps were collected from students. Based on customer feedback, CapChain improved the customer experience by adding QR codes to product packaging, linking customers to the Website and Fanpage for feedback and plant-care support. The project also introduced a personalization service, allowing customers to request custom color combinations for recycled bottle-cap products.

1.1.3.3 Market Expansion Phase (Strategic Roadmap)

- Phase 1: Focus on building a loyal customer base and increasing brand awareness within FPT University Can Tho and surrounding areas.
- Phase 2: Expand distribution coverage across Can Tho City through both offline channels (event booths) and online channels (Fanpage, Website).
- Phase 3: Scale up nationwide via the official e-commerce website capchain.id.vn and major e-commerce platforms (Shopee, TikTok Shop)

- Commercialization and Market Development Phase:

Product Commercialization: Officially launch recycled keychains, mini plant pots, and handmade recycled buttons; refine production processes, pricing, and product packaging.

Distribution Channel Development: Expand sales through websites, Facebook, TikTok, and e-commerce platforms; simultaneously sell directly at schools, events, and trade fairs.

Domestic Market Development: Expand the customer base from students to those who appreciate handmade items, gifts, and a green lifestyle.

Product Portfolio Expansion: Develop additional recycled plastic products such as decorative accessories, handmade gifts, and eco-friendly souvenirs.

Brand Building: Strengthen communication activities about CapChain's recycling story to enhance brand awareness and attract customers.

1.1.3.4 Sustainable Development and Expansion Phase

New Product Research and Development: Continue researching products that can utilize recycled bottle caps and plastic bottles, and further develop the "CapChain Handmade Recycled Buttons" product line.

Overseas Market Development: Gradually access international markets through e-commerce platforms and sales channels for handmade products, gifts, and environmentally friendly products.

Sustainable Development: Expand plastic collection and recycling activities, strengthen cooperation with schools, environmental organizations, and communities.

Improving Product Quality Standards: Refine the raw material processing, manufacturing, and product testing processes to ensure durability, safety, and aesthetics.

Optimizing Production Processes: Gradually improve technology and production processes to increase productivity while maintaining the handmade value and uniqueness of the products.

Building a long-term green brand: Positioning CapChain as a brand of recycled products that are innovative, aesthetically pleasing, and environmentally responsible.

1.2 PRODUCTS

1.2.1 Products Description





The project aims to combine: handmade art, creative design and modern sustainable living trends, to create products that not only have practical value but also contribute to raising community awareness about environmental protection.

CapChain's products are designed with a youthful and modern style, suitable for: students, Gen Z customers, handmade lovers, and people who are interested in sustainable lifestyles.

CapChain focuses on three core handmade recycled product lines that combine artistic creativity, modern aesthetics, and environmental protection messages.

1.2.1.1 CapChain Recycled Handmade Keychain

- Product Name: CapChain Recycled Handmade Keychain.
    - *Product Collections:* The line features four main collections: Alphabet Collection (letters A–Z tailored for personalized initials), Nature Collection (leaf and floral patterns inspired by nature), Character Collection (cute animated characters like Chiikawa), and Mini Art Collection (creative artistic shapes such as hearts and stars).
- Components / Materials:
    - Keychain body: 100% recycled HDPE/PP plastic sourced from collected plastic bottle caps.
    - Connecting ring: Rust-resistant stainless steel key rings for long-term durability.
    - Hanging cord: Handmade woven paracord cord or rustic leather straps.
    - Surface finish: Protective surface coating to improve product longevity and maintain quality.
- Dimensions & Weight:
    - Length: 4 to 7 cm.
    - Thickness: 4 to 8 mm.
    - Weight: Approximately 15 to 40 grams (compact, lightweight, and easy to carry).
- Technical Standards:
    - High water resistance and oxidation resistance, ensuring durability for daily usage.

- Handcrafted with smoothly sanded, polished edges to guarantee absolute safety and comfort for users.
- Signature marble effect: Created naturally through the heat-pressing process, rendering each keychain a 100% unique color pattern.
- Formulated from safe and eco-friendly recycled plastics.

- Product Image:





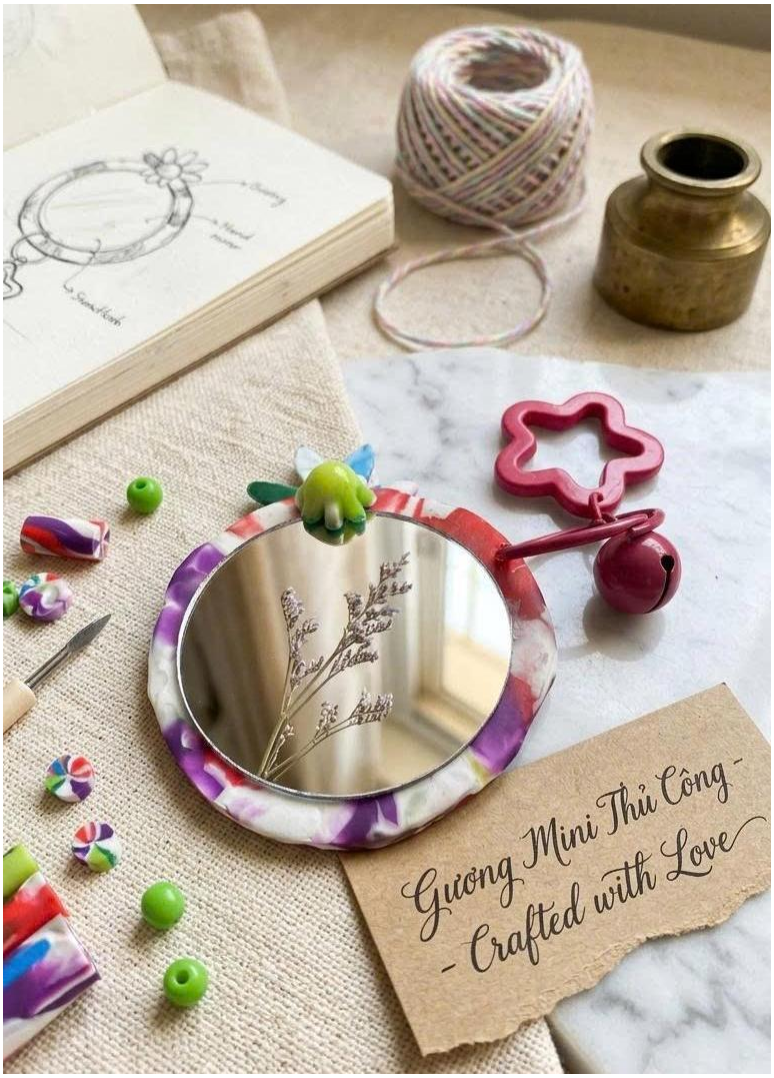
- Packaging & Labeling:
  - *Packaging:* Packed in recycled kraft paper boxes and eco-friendly paper bags to minimize single-use plastic waste.

    - *Brand Tag:* Printed with the CapChain logo, Tagline: *"Creative Green Connection"*, and the environmental commitment: *"This product has saved 5 bottle caps from ending up in landfills"*.
- Functions & Uses:
    - Personal keychains for motorbikes, home keys, and lockers.
    - Trendy fashion charms for backpacks, tote bags, and handbags.
    - Meaningful eco-friendly gifts and souvenirs that spread sustainable living habits.

1.2.1.2 CapChain Recycled Handmade Mirror

- Product Name: CapChain Recycled Handmade Mirror (CapChain Pocket Mirror).
- Components / Materials:
    - Mirror frame: Recycled HDPE/PP plastic from post-consumer bottle caps.
    - Mirror glass: High-quality reflective glass mirror face with clear distortion-free reflection.
    - Decorative accents: Handcrafted bead borders delicately beaded around the frame.
    - Connecting accessories: High-bonding glue and small carrying strap/chain.
- Dimensions & Weight:
    - Diameter: 5 to 8 cm.
    - Thickness: 4 to 8 mm.
    - Weight: Approximately 20 to 60 grams (compact fit for pockets, mini pouches, and hand clutches).
- Technical Standards:
    - Rigid and protective plastic casing that secures the mirror glass from minor shocks.
    - Smooth, carefully polished borders to ensure comfort without sharp edges.
    - Moisture-resistant and easy to clean with a soft cloth.

    - One-of-a-kind marble veining pattern across the frame body.
- Product Image:



- Packaging & Labeling:
    - *Packaging:* Recycled kraft paper gift boxes with cushioned lining, enclosed in a branded kraft bag.
    - *Brand Tag:* CapChain brand tag including a QR code directing to the website and fanpage for customer support.
- Functions & Uses:
    - Personal grooming and quick makeup touch-ups during work or classes.
    - Portable personal accessory or cute bag-hanging ornament.
    - Unique, artisan gift suitable for birthdays and special celebrations.

1.2.1.3 CapChain Recycled Mini Plant Pot

- Product Name: CapChain Recycled Mini Plant Pot (CapChain Plant Pot).
- Components / Materials:
    - Pot body: 100% recycled plastic upcycled from post-consumer PET plastic water bottles.
    - Live plant: Real succulents or mini cacti, handpicked for strong resilience and vitality.
    - Soil mix: Nutrient-dense and well-draining potting soil formulated for succulents.
    - Top layer: Decorative colored pebbles (white, green, or natural pebbles) to stabilize the plant base and retain soil moisture.
    - Accents: Integrated curved plastic handle and rustic cord.
- Dimensions: Available in three flexible sizes:
    - *Small:* 7 x 7 x 10 cm.
    - *Medium:* 8 x 8 x 11 cm.
    - *Large:* 10 x 10 x 14 cm.
- Technical Specifications / Criteria:
    - Form-factor: Styled like a miniature basket with an ergonomic curved handle and petal-shaped trimmed rim.
    - Color tones: Natural clear PET plastic, subtle transparent blue, or fresh mint green.
    - Drainage: Engineered with bottom drainage holes to prevent root rot in indoor conditions.
    - Lightweight and shatterproof, offering superior drop resistance compared to ceramic pots.
- Product Image:



- Packaging & Labeling:
    - *Packaging:* Breathable recycled kraft paper boxes with transparent display windows and branded paper carrier bags.
    - *Brand Tag & Add-ons:* Kraft tag printed with the circular recycling symbol and CapChain branding; accompanied by an instructional leaflet on succulent care.
    - *Core Brand Message: "Giving Plastic a New Life – Building a Greener Future".*
- Functions & Uses:
    - Desk plant for greening study spaces, home offices, and bookshelves.
    - Natural eye relaxation and stress relief during demanding study sessions or deadline rushes.
    - Educational green gift encouraging recycling habits and nature connection.

1.2.1.4 CapChain Recycled Fashion Buttons

- Product Name: CapChain Handmade Recycled Fashion Buttons (CapChain Eco-Buttons / Recycled Fashion Buttons).
- Product Collections / Variations:

    - Classic Round Collection: Features traditional circular designs (available in 2-hole or 4-hole configurations) with minimalist, elegant marble textures.
    - Artistic Shapes Collection: Features stylized designs such as four-petal flowers, hexagons, or hearts shapes, specifically crafted for DIY and upcycling fashion enthusiasts.
- Materials / Ingredients:
    - Button Body: 100% recycled HDPE/PP plastic sourced from collected plastic bottle caps, meticulously color-sorted and heat-processed.
    - Surface Finish: Hand-polished with sandpaper and specialized plastic polishing compound to achieve a smooth, glossy, and refined surface without the use of harmful chemical coatings.
- Product Dimensions & Weight:
    - Button Diameter: Diverse sizing options:
        - *Small (15 mm):* Ideal for dress shirts, blouses, and collars.
        - *Medium (20 mm):* Suitable for knitwear, sweaters, and cardigans.
        - *Large (25 mm):* Designed for blazers, coats, canvas tote bags, and backpacks.
    - Thickness: 2.5 – 4 mm (ensuring structural rigidity and fastening durability).
    - Weight: Ultra-lightweight, approximately 1 – 3 grams per button (around 10 – 15 grams per set of 4–6 pieces).
- Technical Standards:
    - Heat & Wash Resistance: High-grade HDPE/PP plastic exhibits superior elasticity and heat resistance; safe for machine washing and detergent soaking without color fading, cracking, or becoming brittle over time.
    - Mechanical Precision: Thread holes are neatly drilled and chamfered with completely smooth edges, free of sharp burrs, preventing thread fraying or fabric scratching during daily wear.

- One-of-a-kind Marble Texture: Colors and swirling patterns are formed naturally through the thermal blending of assorted plastic flakes, ensuring that every single button is truly unique.

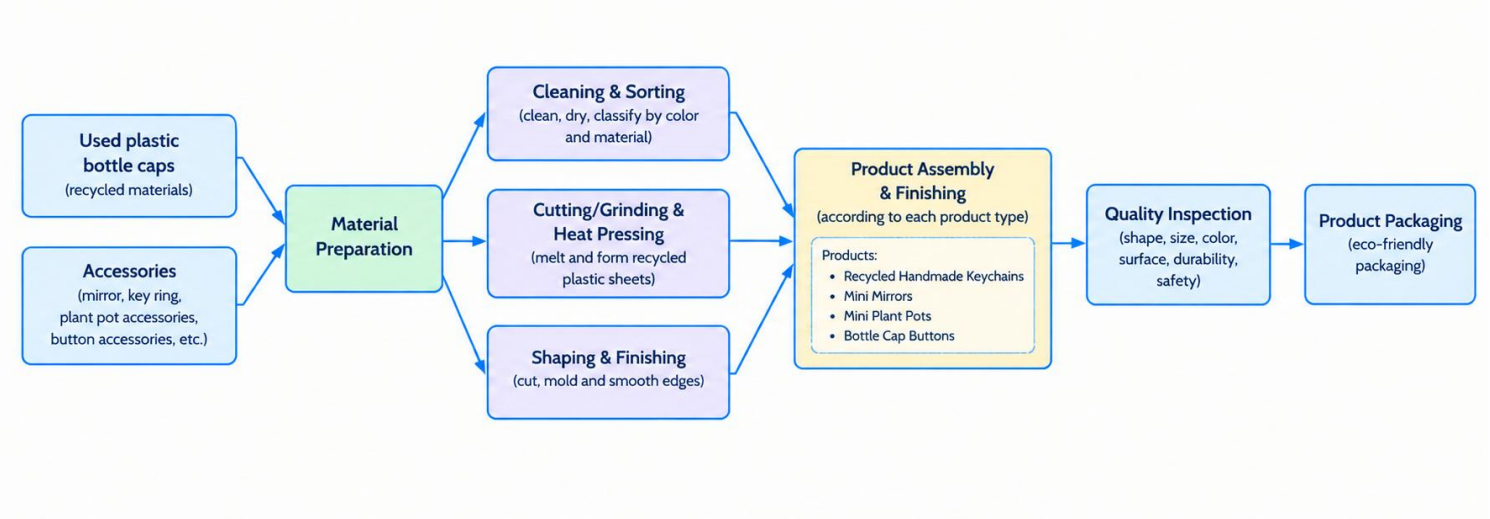
- Product Image:



- Packaging & Labeling:
  - Packaging: Mounted neatly in sets (Set of 4 or Set of 6) on sturdy recycled Kraft paper backing cards, encased in an eco-friendly frosted glassine paper sleeve.
  - Label: Debossed with the CapChain logo and the campaign slogan *"Small Buttons – Big Green Impact"*; the back includes a QR code linking to styling videos and DIY clothing refashion/upcycling tutorials.
- Functions & Uses:
  - Serves as functional replacements or stylish upgrades for buttons on dress shirts, sweaters, cardigans, and vintage/rustic-styled blazers.
  - Functions as creative decorative accents on canvas tote bags, bucket hats, backpack webbing, or handmade notebook covers.
  - Caters to the DIY and upcycled fashion community, turning thrifted or pre-loved garments into innovative, personalized fashion statements.

- Makes for a charming, delicate eco-friendly gift and souvenir that spreads the message of sustainable consumption.

1.2.2 Description of the production process for the product



Detailed Description of the Production Process

The production process of CapChain begins with the preparation of materials. Used plastic bottle caps are collected and inspected, while the necessary accessories for each product, such as key rings, mirror components, plant pot accessories, and button accessories, are prepared. This ensures that all materials are ready before production begins.

Next, the bottle caps undergo cleaning and sorting. They are cleaned, dried, and sorted according to their colors and plastic types. After that, the bottle caps are cut or ground into small pieces and heat-pressed to form recycled plastic sheets. These sheets are then shaped according to the specific design of each product. The edges and surfaces are carefully sanded and smoothed to improve both the appearance and safety of the finished products.

The next stage is product assembly and finishing. The recycled plastic components are combined with the corresponding accessories to create different products, including keychains, mini mirrors, mini plant pots, and bottle-cap buttons. Each product is completed according to its specific function and design.

After assembly, all finished products go through quality inspection. The products are checked for shape, size, color, surface quality, durability, and safety.

Products that do not meet the required standards are either reworked or removed from the production line.

Finally, products that pass the quality inspection are packaged using environmentally friendly materials. This packaging approach is consistent with CapChain's focus on recycling and sustainable development, ensuring that the entire production process reflects the brand's environmental orientation.

1.2.3 Production Capacity

CapChain's production capacity is supported by a six-member team, recycled plastic materials collected from schools and the community, and the tools and equipment required for cleaning, cutting, heat pressing, shaping, and product finishing. The production process is carried out directly by the team to maintain control over product design and finished-product quality.

For keychains and mini mirrors, plastic bottle caps are cleaned, sorted by color, cut into small pieces, heat-pressed, shaped, and finished. For mini plant pots, used plastic bottles are cleaned, cut, shaped, hand-decorated, and combined with cactus plants.

With the current handmade production model, CapChain estimates that it can complete approximately 40–50 products per day, depending on the product type, design complexity, and availability of raw materials.

1.2.4 Quality

CapChain's product quality is based on durability, safety, aesthetics, and practical usability. This is an important factor because the market survey shows that 20.8% of respondents are dissatisfied with the durability of existing recycled products.

1.2.4.1 Product Quality Standards

Criteria	Requirements
Raw Materials	Recycled plastic is collected, sorted, and cleaned before production.
Durability	Products have a solid structure and are suitable for daily use.

Safety	Product edges and surfaces are smoothed to minimize sharp edges.
Aesthetics	Products have clear shapes, colors consistent with the design, and clean, well-finished surfaces.
Accessories	Keychains use rust-resistant metal rings; mirrors have safe finished edges; plant pots have suitable drainage holes.
Packaging	Kraft paper boxes, paper bags, and brand labels are used in line with the project's eco-friendly direction.

1.2.4.2 Product Quality Control Process



Quality Control Process

The quality control process at CapChain consists of five main stages to ensure that recycled materials and finished products meet the required standards.

Step 1 is input material inspection, in which collected bottle caps and plastic bottles are inspected, sorted, and cleaned. Any materials that are unsuitable for production are removed.

Step 2 is post-cleaning inspection, where the materials are checked to ensure that they are completely clean and dry before being processed through cutting, heat pressing, or shaping.

Step 3 is in-process inspection. During production, the team monitors important stages such as cutting, material arrangement, heat pressing, molding, and shaping. This helps minimize defects related to the product's shape, surface, and color.

Step 4 is finished product inspection, during which the final products are checked for their shape, surface, edges, accessories, and usability. Products that do not meet the requirements are either corrected or removed.

Finally, Step 5 is pre-packaging inspection. Products that pass the quality inspection are cleaned once again and packaged using paper-based packaging. This packaging method is consistent with CapChain's eco-friendly direction and helps maintain the brand's commitment to recycled and sustainable products.

APPENDIX

*** ASSESSING TEAM PERFORMANCE**

Evaluation criteria will be based on:

- Progress in finishing the work
- A willingness to contribute ideas and goodwill to group projects
- Plan and work together with team members
- Establish connections and communicate clearly with team members

No.	Members	Student Code	Job description	Score 1 (Outcome 1)	Score 2 (Outcome 2)	Score 3 (Outcome 3)	Score (Full Process)	Note
1	Nguyễn Ái Nhân (Leader)	CS181073	Project planning, team coordination, and fanpage management	100%				
2	Nguyễn Thị Ngọc Hân	CA180407	Budget planning and financial management	100%				

3	Lê Minh Kha	CE182100	System development, technical support, and maintenance	100%				
4	Trần Thị Hồng Nhung	CE191707	Branding design and product design	100%				
5	Ngô Thành Danh	CE191720	Customer relationship management and sales coordination	100%				
6	Lê Ngọc Trâm Anh	CS191584	Marketing strategy, market research, and campaign planning	100%				